SRI MANAKULA VINAYAGAR
ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)
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(ACCREDITED BY NBA-AICTE; NEW DELHI, ACCREDITED BY NAAC WITH "A" GRADE)
MADAGADIPET, PUDUCHERRY - 605 107

Batch : CSE 04

ProAuth IQ

Prior Authorization Triage and Policy Companion

cognizant

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PROBLEM STATEMENT

Prior authorization requires insurance teams to review **patient information, diagnosis, clinical evidence, supporting documents, and coverage policies** before approving medical services. Existing processes often rely on **manual review and predefined rules**, making them difficult to manage when policies are complex, information is unstructured, or documents are incomplete.

This can lead to **increased administrative effort, inconsistent evaluation, and delays** for providers and patients.

Therefore, there is a need for a system that can **evaluate requests against applicable coverage requirements and clinical evidence while providing transparent reasoning**.

The system should support three outcomes:

- **Approve** – Requirements are sufficiently satisfied.
- **Pend for Nurse Review** – Further clinical assessment is required.
- **Request More Information** – Required information or evidence is missing.

The evaluation should also maintain an **audit trail** to ensure transparency and traceability.

OUR SOLUTION

We propose **ProAuth IQ**, an AI-powered prior authorization system that combines **Policy RAG, agentic AI, and machine learning** to streamline the evaluation of healthcare authorization requests.

Doctors submit **patient details, diagnosis, requested medical service, insurance information, clinical justification, and supporting documents** through the system. The application identifies the applicable Medicare coverage context and uses **Retrieval-Augmented Generation (RAG)** to retrieve relevant policy sections, coverage criteria, and requirements.

A **Lang-Graph-based agentic workflow** coordinates specialized agents to analyse different aspects of the request. The **Policy Agent** retrieves and interprets applicable coverage requirements, the **Clinical Agent** analyses the diagnosis and clinical context, and the **Document Analysis Agent** examines uploaded documents to identify supporting or missing evidence. Their outputs are combined by the reasoning layer into structured evaluation criteria.

The resulting evidence is converted into **structured ML features**, such as policy compliance, documentation completeness, clinical evidence availability, missing criteria, and policy relevance. These features are provided to the **ML triage model**, which recommends one of three outcomes:

- **Approve** – the available evidence sufficiently satisfies the applicable requirements.
- **Pend for Nurse Review** – the request contains uncertainty or requires further clinical assessment.
- **Request More Information** – required information or supporting evidence is missing.

The system also provides an **explainable recommendation** rather than only displaying a prediction. Users can view the relevant policy criteria, supporting evidence, missing information, evaluation results, and prediction confidence.

Finally, important authorization activities, policy references, evaluation results, model predictions, and workflow events are maintained through an **audit trail**, providing transparency, traceability, and a clear record of how each authorization request was evaluated.

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PROJECT CHALLENGES

1. Complex and Variable Coverage Policies

Insurance policies contain detailed coverage requirements, conditions, documentation criteria, limitations, and exceptions. Identifying the applicable policy and relevant sections is challenging.

Our approach: We use a Policy RAG pipeline to retrieve the most relevant policy sections and coverage criteria instead of depending on static hardcoded rules.

2. Connecting Clinical Information with Policy Requirements

Patient diagnosis, requested service, and clinical evidence need to be evaluated together against the applicable coverage criteria.

Our approach: The Clinical Agent and Coverage Reasoning Agent structure the clinical information and compare it with the retrieved policy requirements to identify supported, missing, or uncertain criteria.

3. Processing Unstructured Medical Documents

Supporting documents such as clinical notes and reports may contain important evidence that is not available in structured patient data.

Our approach: The Document Analysis Agent, supported by OCR/text extraction, analyzes uploaded documents to identify relevant evidence and missing information.

4. Reliable and Grounded AI Reasoning

LLMs can generate unsupported information if they are not properly constrained, which is especially problematic in healthcare workflows.

Our approach: Agents are grounded using retrieved policy evidence and submitted clinical information and are instructed to identify information as supported, missing, or unknown rather than inventing facts.

5. Integrating Agentic AI with ML Triage

Agents are useful for extracting and reasoning over evidence, while the ML model requires consistent structured inputs for prediction.

Our approach: The agents convert the collected evidence into structured features, which are passed to the ML model for triage into Approve, Pend for Nurse Review, or Request More Information.

FEEDBACK FROM MENTOR

1. Implement Policy RAG

The mentor suggested implementing a Retrieval-Augmented Generation (RAG) approach for insurance policy evaluation. Instead of depending only on predefined or hardcoded rules, the system should retrieve the relevant sections directly from Medicare coverage policy documents based on the patient's diagnosis and requested medical service.

Our Approach:

We implement a Policy RAG pipeline that converts policy documents into searchable chunks and uses semantic retrieval to identify the most relevant policy sections for each authorization request. The retrieved policy evidence is then provided to the Policy Agent through the LangGraph workflow, allowing the system to base its evaluation on the actual policy content.

2. Improve the Policy Checking System

The existing policy-checking approach relied heavily on static rules, which makes it difficult to adapt when coverage requirements change or when new policies are introduced.

Our Approach:

We move from fixed hardcoded rules toward a configurable, policy-driven evaluation system. The retrieved policy is used to identify applicable coverage criteria, medical-necessity requirements, and documentation requirements. The Coverage Reasoning Agent then evaluates these criteria against the submitted request. This allows new policy documents to be added to the knowledge base without changing the core application logic.

3. Improve Evidence-Based Reasoning

The system needs to establish a clear connection between the patient's information, diagnosis, requested service, supporting documents, and applicable insurance policy. Simply matching individual fields is not sufficient when evidence is distributed across multiple documents and policy sections.

Our Approach:

We use specialized Clinical and Document Analysis Agents to extract relevant evidence from the request and uploaded documents. The retrieved policy criteria are then combined with this evidence by the Coverage Reasoning Agent, which categorizes each requirement as Satisfied, Missing, or Uncertain. These structured results are passed to the ML model as meaningful features, improving the basis for the final triage recommendation and making the result easier to explain.

TEAM MEMBERS AND ROLES

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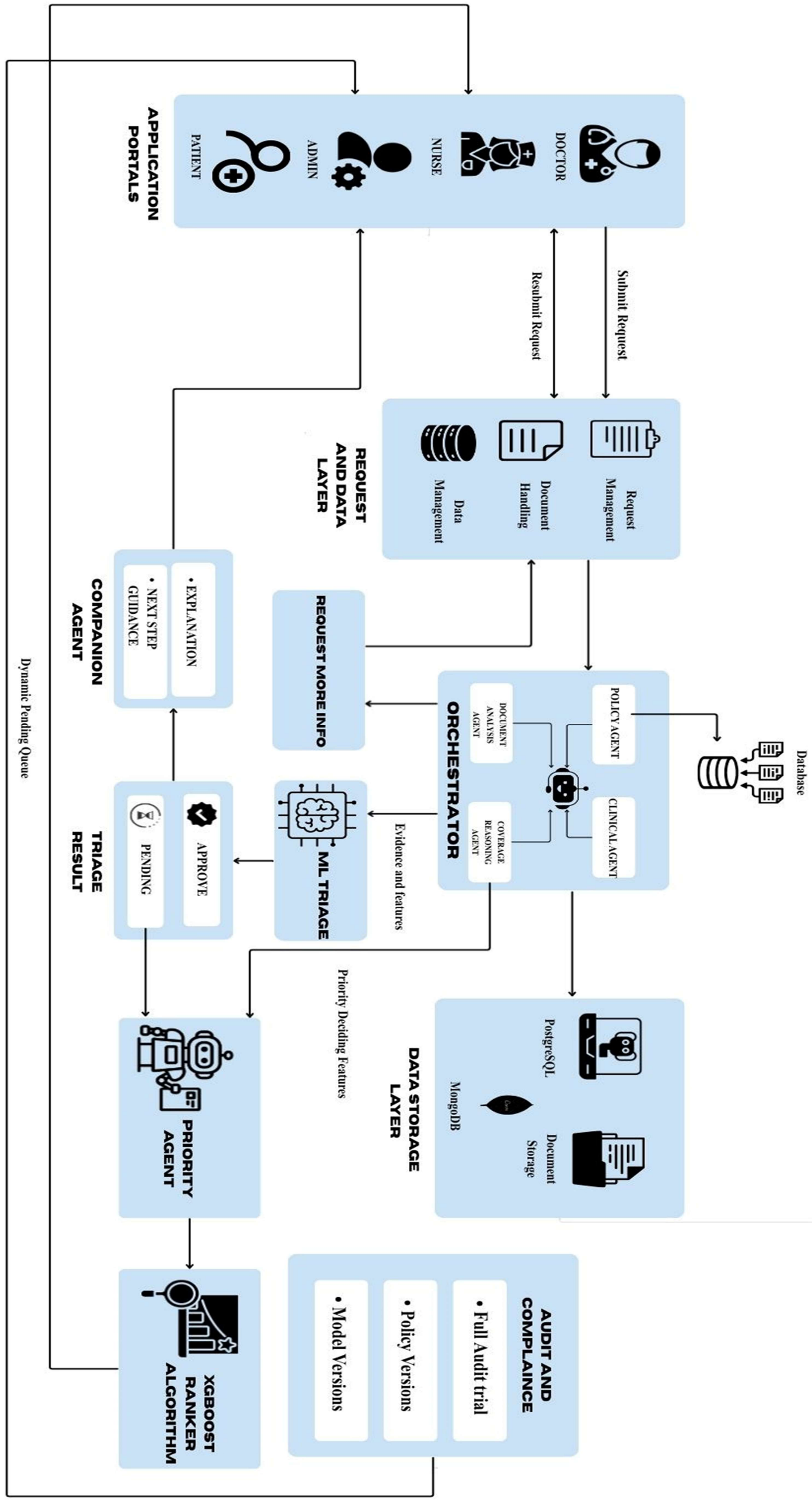
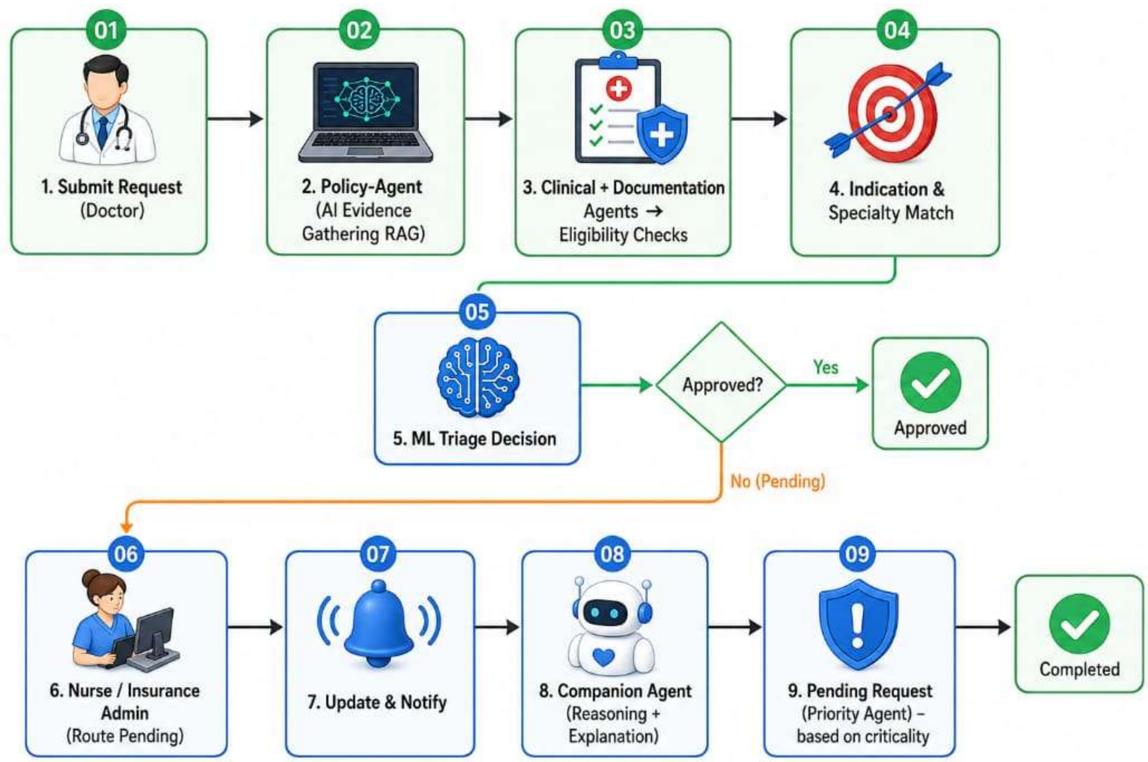
ARCHITECTURE DIAGRAM



MODEL FLOW DIAGRAM



LINKS

Cloud Link: **http://proaut-Publi-o9CFLIlXCkqI-540122485.ap-south-1.elb.amazonaws.com**

GitHub Link: **https://github.com/Abdullah-218/Prior-Authorization-NPN**

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